

INVENTION TITLE

SYSTEM AND METHOD FOR DISTRIBUTED GENERAL-PURPOSE COMPUTING WITH CRYPTOTOKEN PAYMENT SYSTEM

FIELD OF THE INVENTION

[0001] The invention relates generally to system and method for Distributed General-Purpose Computing with a Cryptotoken Payment System

BACKGROUND OF THE INVENTION

[0002] Currently there are a number of solutions for high speed processing of transactions via a Blockchain Cryptotoken. Some of these solutions attempt to use different techniques to speed up the verification of transactions, but these solutions fail to meet the needs of the industry because they current systems verification of transactions take several seconds to several minutes. Other solutions attempt to solve this issue by using Directed Acyclic Graph Blockchain Cryptotokens, but these solutions are similarly unable to meet the needs of the industry because they aren't integrated with decentralized applications.

SUMMARY OF THE INVENTION

[0003] A system and method for distributed general-purpose computing with cryptotoken payment system provides a full system that integrates a slow Blockchain Cryptotoken with a fast Directed Acyclic Graph Blockchain Cryptotoken. The instant invention provides: (1) a hybrid Cryptocurrency/Cryptotoken system that allows for very fast transactions to happen decoupled from the slow transactions of normal Cryptocurrencies (2) a mechanism to verify the processing of data and payments to the End User's wallet for the processing of that data that can in turn be used for In App purchases or in game purchases of items and (3) mechanisms for the transfer of unprocessed and processed data to the client or customer.

[0004] It is desirable to have a system architecture with associated software that allows for the distribution of payments for the processing of data that utilizes unused cycles of Compute Devices and Chips. Furthermore, it is desirable to have a system and software that can be easily integrated into computer and mobile applications and games. Still further, it is desirable to have a system and software that allows the payments made to be converted to currency or in-app purchases. The disclosed system and associated software advantageously fill these needs and addresses the aforementioned deficiencies by providing a set of

components to interface to computer and mobile devices apps and games to access a Cryptotoken and the compute devices.

[0005] Disclosed is a system together with an associated computer process. The system is made up of the following components: A Blockchain Cryptotoken component, Data Delivery/Storage component and a Processing component.

[0006] These components are connected as follows: The Blockchain Cryptotoken is used to receive payment for the processing of data. The Data Delivery/Storage component is used to store data before and after processing of the data. The Processing Component is used to process and return the processed data to the Data Delivery/Storage component. Once the data has been processed the Blockchain Cryptotoken is awarded to the device owner that has done the processing.

[0007] The associated computer process is made up of the following executable steps, all of which are required in all versions: A customer sends or converts currency which is converted into Cryptotokens. The customer then sends the data that is to be processed to the Decentralized File System component and it is stored in the Delivery/Storage component.

[0008] The Processing Component then processes the data and returns the data to the Delivery/Storage Component. Upon successful verification of the work being performed, the Blockchain Cryptotoken releases the Cryptotokens to the End User's Wallet that processed data.

[0009] The system may also have one or more of the following optional components: Decentralized Website Component, which is a user facing input system, such as a decentralized website with web forms that allows a customer to interface to purchase processing of data.

[0010] The disclosed system is unique when compared with other known systems and solutions in that it provides a full system that integrates a slow Blockchain Cryptotoken with a fast Directed Acyclic Graph Blockchain Cryptotoken. Similarly, the software disclosed is unique when compared with other known solutions in that it provides a unique way for applications and games to utilize unused cycles from computing devices to earn Cryptotokens by processing real data, such as Artificial Intelligence or Machine Learning data. Typically, other Cryptocurrencies and Cryptotokens are earned by what is known as mining. Mining uses the computing devices to solve complex mathematical problems to verify a block on the blockchain. This mining does not actually process real data and is only used as a competition to see who wins the mining payments.

[0011] The disclosed software is unique in that the overall architecture of the system is different from other known systems. More specifically, it provides: (1) a hybrid Cryptocurrency/Cryptotoken system that allows for very fast transactions to happen decoupled from the slow transactions of normal Cryptocurrencies (2) It provides for a mechanism to verify the processing of data and payments to the End User's wallet for

the processing of that data that can in turn be used for In App purchases or in game purchases of items and (3) It provides mechanisms for the transfer of unprocessed and processed data to the client or customer. Similarly, the associated software is unique in that it incorporates interfaces that allow for applications and games to use unused cycles of one or more devices to process real data and to receive and utilize payments for the processing of data.

[0012] This disclosure will now provide a more detailed and specific description that will refer to the accompanying drawings. The drawings and specific descriptions of the drawings, as well as any specific or alternative embodiments discussed, are intended to be read in conjunction with the entirety of this disclosure. The System and Method for Distributed General-Purpose Computing with CryptoToken Payment System may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided by way of illustration only and so that this disclosure will be thorough, complete and fully convey understanding to those skilled in the art.

[0013] Other features and advantages of the instant invention will become apparent from the following description of the invention which refers to the accompanying drawings.

[0014]

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] FIG. 1 is a schematic diagram of a distributed general-purpose computing system with cryptotoken payment system according to an embodiment of the invention.

[0016] FIG. 2 is a schematic diagram of the internal components of a Software Development Kit.

[0017] FIG. 3 is a flow diagram of a typical client or customer transaction.

[0018] FIG. 4 is a flow diagram of the internal workings of the processing request inside a Decentralized App.

[0019] FIG. 5 is a flow diagram of how an Application or Game (App/Game) might use the Cryptotokens for in-app or in game purchases.

[0020] FIG. 6 is a flow diagram of the processing of data flow.

[0021] FIG. 7 is a flow diagram of the verification step that all data was really processed.

[0022] FIG. 8 is a flow diagram of the verification step that verifies all blocks of a transaction request to process data.

DETAILED DESCRIPTION OF THE INVENTION

[0023] In the following detailed description of the invention, reference is made to the drawings in which reference numerals refer to

like elements, and which are intended to show by way of illustration specific embodiments in which the invention may be practiced. It is understood that other embodiments may be utilized and that structural changes may be made without departing from the scope and spirit of the invention.

[0024] The present invention is directed to system and method for Distributed General-Purpose Computing with Cryptotoken Payment System.

[0025] Disclosed is a system together with an associated computer process. The system is made up of the following components: Generic Blockchain Cryptotoken component, Directed Acyclic Graph Blockchain Cryptotoken component, Payment System component, Proof of Work component, Decentralized File System component, Data Delivery/Storage component, Processing component, Communications component, Software Development Kit component, Cryptocurrency Wallet component and Compute Devices.

[0026] These components are connected as follows: The Generic Blockchain Cryptotoken component interfaces with a external Currency Converter system and is connected to the Proof of Work component and the Directed Acyclic Graph Blockchain Cryptotoken component. The Proof of Work component is connected to the Communication component. The Directed Acyclic Graph Blockchain component is connected to the Communication component. The Communication component is

connected to the Directed Acyclic Graph Blockchain Cryptotoken component, the Cryptotoken Wallet component and the Software Development Kit component. The Decentralized File System Component interfaces with data to process and is connected to the Delivery/Storage component. The Cryptotoken Wallet component is connected to the Communications component and the Software Development Kit component and interfaces with external App/Game systems. The Delivery/Storage component is connected to the Software Development Kit component and interfaces with external Compute Devices. The processing component interfaces with Compute Devices and is connected to the Software Development Kit component.

[0027] The associated computer process is made up of the following executable steps, all of which are required in all versions: A customer pays currency into a converter system that converts into a Cryptotoken and sends the transaction to the Generic Blockchain Cryptotoken component which holds the payment as a deposit. The customer then sends the data that is to be processed to the Decentralized File System component and it is stored in the Delivery/Storage component.

[0028] The customer then initiates a processing request with the Decentralized File System components location and the Generic Blockchain Cryptotoken deposit account. The Generic Blockchain Cryptotoken component calculates out the cost of the processing and puts that amount of Cryptotoken on hold for processing. The Generic

Blockchain Cryptotoken component then deposits into the Directed Acyclic Graph Blockchain component equivalent amounts of the Directed Acyclic Graph Blockchain Cryptotoken into the Directed Acyclic Blockchain. The Directed Acyclic Graph Blockchain Cryptotoken component then sends the processing request to the Communication component that distributes the work to a multitude of End User Devices for processing through the Software Development Kit.

[0029] The Software Development Kit component then sends the request to the Processing component. The processing component then communicates with Compute Devices to process the data. The Compute Devices process the data and sent the processed data to the Delivery/Storage component and signals back to the Processing component the completion of the processing. The processing component then signals back to the Software Development Kit component that the processing is completed. The Software Development Kit component then signals back to the Communication component that processing is done for the one device.

[0030] When all processing is reported back to the Communication component it sends a signal back to the Proof of Work component that verifies the integrity of the processed data and then signals the Generic Blockchain Cryptotoken component to release the hold on the Generic Blockchain Cryptotoken. The Generic Cryptotoken then sends a signal to

the Directed Acyclic Graph Blockchain token to release Cryptotokens to the End User's Cryptotoken Wallet.

[0031] FIG. 1 is a schematic diagram that illustrates a Distributed General-Purpose Computing System with Cryptotoken Payment System 1000, according to an embodiment. In some embodiments, a Distributed General-Purpose Computing System with Crypto-Token Payment System can also be referred to as, for example, a Cryptocurrency payment system for General Compute Devices, a Decentralized Crypto Payment system for Computing, and/or the like.

[0032] As shown in FIG. 1, the Distributed General-Purpose Computing System 1000 includes a Decentralized Application 1002, an End User's Device 1006 and a Directed Acyclic Graph Blockchain Cryptotoken Component 1004. The Decentralized Application 1002 is in communication with the End User's Device 1006 and the Directed Acyclic Graph Blockchain Cryptotoken 1004 through the Communication Component 1005 and via a network 1015. The network 1015 can be any type of network (e.g., a local area network (LAN), a wide area network (WAN), a virtual network, a telecommunications network, a cellular network, a wireless LAN (WLAN), etc.) configured to enable communication between the Decentralized Application 1002, the Directed Acyclic Graph Blockchain Cryptotoken component 1004 and the End User's Device 1006. The network 1015 can be implemented as a wired

network and/or wireless network, or any other communication such as Bluetooth, NFC, etc.

[0033] The Decentralized Application 1002 contains three components, the Generic Blockchain Cryptotoken component 1003, the Communication Component 1005 and the Proof of Work component 1009. The End User's Device 1006 has six components, the Decentralized File System 1008, the Cryptotoken Wallet 1012, the Software Development Kit API 1013, the Delivery/Storage component 1009 and the Processing component 1010. The Directed Acyclic Graph Blockchain Cryptotoken 1004 is a stand-alone component.

[0034] The Generic Blockchain Cryptotoken component 1003 receives a Processing/Currency Conversion Request 1001 from an outside system. The Processing/Currency Conversion Request 1001 can be initiated from any number of systems, including Web Applications, other Decentralized Applications (DApps), Computer Applications and any other systems that can interface to a Generic Blockchain Cryptotoken system. A Currency Conversion Request 1001 is communicated to the Decentralized Application 1002 and the Generic Blockchain Cryptotoken component 1003. Data to be processed is also communicated the End User's Device 1006 and the Decentralized File System 1008. The Currency Conversion Request 1001 and the Data to Process 1007 communication may be done in parallel.

[0035] One or more End User's Device receives data to be processed via an outside system. The Data to Process 1007 can be initiated from any number of systems including Web Application uploads, File System transfers via ftp, or even a simple URL pointer to the data. The Decentralized File System 1008 component then stores that data in one or more End User's Devices 1006 for access.

[0036] The App/Game Interface 1014 is an external system that communicates with the Software Development Kit API 1013. The App/Game Interface 1014 external system communicates a hybrid set of compute instructions to the Software Development Kit API 1013. One embodiment of the hybrid set of compute instructions are called Shaders. Other embodiments include CPU instructions, ASIC instructions, etc. and are dependent on the type of Compute Device being utilized and the instructions that are native to that Compute Device and how the commands are communicated to the Compute Devices. The Software Development Kit API 1013 then forwards the hybrid set of compute instructions to the Processing component 1010, which stores the for later use.

[0037] After the Currency Conversion Request and the Data to Process is communicated a processing request is made to the Decentralized Application 1002 and the Generic Blockchain Cryptotoken component 1003. The request contains the address of the data to be processed that was sent the Decentralized File System 1008. The Generic

Blockchain Cryptotoken component 1003 then uses the Communication Component to communicate to the Directed Acyclic Graph Blockchain Cryptotoken 1004 component to instantiate Cryptotokens and then signals the Software Development Kit API 1013 to start processing the request.

[0038] Upon the Software Development Kit API 1013 receiving the request for processing, Software Development Kit API 1013 Delivery/Storage Component 1009 to transfer the data to be processed from the Decentralized File System 1008 to one or more Compute Devices 1011. The Software Development Kit API 1013 then signals the Processing component 1010 that processing is ready. The Processing Component 1010 then instantiates the hybrid set of compute instructions inside one or more Compute Devices 1011.

[0039] The App/Game Interface 1014 sends a signal to the Software Development Kit API 1013 that the current data has been processed. The App/Game Interface 1014 can signal with events such as end of frame while running a game or end of processing while running an application. These events signal the Software Development Kit API 1013 to message the Delivery/Storage Component 1009 to extract processed data from one or more Compute Devices 1011. The Delivery/Storage Component 1009 then places the processed data in the Decentralized File System 1008. The Software Development Kit API 1013 then communicates through the Communication Component 1005 to the Proof of Work

Component 1009 that the processing is completed and the address in the Decentralized File System 1008 of the processed data.

[0040] Upon receiving one or more data processed messages The Proof of Work Component 1009 verifies the data was actually processed correctly. Once the Proof of Work Component 1009 verifies that the data is valid for one or more End User's Devices 1006 involved in the processing transaction it sends a message through the Communication Component 1005 to the Generic Blockchain Cryptotoken 1003 to release the escrow in the contract. The Generic Blockchain Cryptotoken 1003 then marks the associated transaction complete and releases the Generic Blockchain Cryptotokens to the End User's Device Cryptotoken Wallet 1012. The Generic Blockchain Cryptotoken 1003 then communicates through the Communication Component 1005 to the Directed Acyclic Graph Blockchain Cryptotoken 1004 that processing is finished so that it can do cleanup or pruning of the blockchain.

[0041] FIG. 2 is a schematic diagram of the internal components of the Software Development Kit 2001 and the Software Development Kit API 1013. The General-Purpose Compute Devices 2002 are a representation of any type of computing device. These can include Graphics Processing Units (GPU(s)) 2006, Central Processing Units (CPU(s)) 2007 and other types Processing Units 2008. A Device App/Game 2000 is connected to the Software Development Kit 2001 via an Application Programming Interface (API) 2009, 2010.

[0042] The first interface 2009 is the interface to the Cryptotoken Deposit/Withdrawal 2004 component. This allows the Device App/Game 2000 the ability to deposit and withdraw Cryptotokens to the external component Cryptotoken Wallet 2013.

[0043] The second interface 2010 is the interface that the App/Game uses to setup and execute processing of data. The Processing Component 2005 communicates to the Delivery Storage Component 2011 which retrieves data from the Decentralized File System 2003 component. This data was previously sent by a client or customer via an outside system, Data to Process 2012. The Processing Component 2005 then sends a hybrid instruction set to the General-Purpose Computing Devices 2002 that are available for processing. The Processing Component 2005 also transfers the data to be processed between the General-Purpose Computing Devices 2002. The Device App/Game 2000 then executes its normal process which in executes the hybrid instruction set that processes the data.

[0044] At set intervals the Device App/Game 2000 signals to the Processing Component that the data has been processed. The Processing Component 2005 then retrieves the process data from the General-Purpose Computing Devices 2002 and sends the data to the Delivery/Storage Component 2011. The Delivery/Storage Component 2011 then delivers the processed data to the client or customer via the Decentralized File System 2003.

[0045] FIG. 3 is a flow diagram of a typical client or customer transaction. A customer or client converts any currency to a Cryptotoken 3000. This token is sent to the Generic Blockchain 3001.

[0046] The Currency Conversion step 3000 is communicated to the Generic Blockchain Cryptotoken component 3001. The Generic Blockchain Cryptotoken component 3001 verifies payment via normal blockchain transaction verification systems. The Generic Blockchain Cryptotoken component 3001 then stores or adds to the balance in what is known as a contract, inside the Cryptotoken component 3001 within the address of a transaction block of a blockchain.

[0047] The client or customer then sends the data to be processed to the Decentralized File System 3002, 3003. This step and the Currency Conversion step 3000 can be done synchronously or asynchronously. The system then deposits the Cryptotokens into the into a specific transaction ID for the customer/client 3004. The customer then sends the address of the data to be processed and a request to process the data to the Generic Blockchain component 3005. The processing of data can be done asynchronously or synchronously 3006. If the processing is synchronous and Immediate response is returned 3007. In this system asynchronous process of the data will be the most common resulting in a delay for the signal that the data was processed 3008.

[0048] The Processing Request 3005 is made to the Generic Blockchain Cryptotoken component 3001. The Generic Blockchain

Cryptotoken component 3001 then calculates the amount of Cryptotokens that are necessary to do the processing request and if that amount is available in the contract, it moves that amount of Cryptotokens into an escrow entry within the blockchain and assigns a new transaction ID for that escrow entry.

[0049] FIG. 4 is a flow diagram of the internal workings of the processing request inside the Decentralized App (DApp). The DApp calculates the amount of Cryptotokens required for this transaction 4001. The calculation uses the amount of processing power needed and the size of the data to process in determining the amount. If there is enough Cryptotoken on deposit to execute this transaction 4002. If there is sufficient Cryptotokens available, the Cryptotokens are deposited into the Directed Acyclic Graph Blockchain 4003. The DApp then sends the processing request to the Software Development Kit API 4004. If there are not enough Cryptotokens the request, then is aborted and an error will be returned 4005.

[0050] FIG. 5 is a flow diagram of how an Application or Game (App/Game) might use the Cryptotokens for in-app or in game purchases. The App/Game first checks the wallet of the user or play to see if there are sufficient Cryptotokens for the transaction 5000. If there are enough Cryptotokens, a withdrawal from the Wallet is made 5003. If there are not enough Cryptotokens the user can add Cryptotokens by making an in-app or in game purchase that will deposit a certain

percentage into the Generic Blockchain 5001 and then add equal amounts to the Directed Acyclic Cryptotokens 5002 via a Cryptotoken wallet and then rechecking the amount of Cryptotokens.

[0051] FIG. 6 is a flow diagram of the processing of data flow. The first check is to make sure the devices have compute devices available for processing of data. If not, the flow ends immediately. If it does the process retrieves the transaction ID from the Directed Acyclic Graph Blockchain system 6002. The process then retrieves the data from the Delivery/Storage Component 6003. The process then adds the data to the Compute Devices for processing 6004.

[0052] The App/Game then sends the hybrid instruction set to the Software Development Kit API which enables the Combined Compute Instructions 6005. When the App/Game does its normal processing, the hybrid Compute Instructions are executed 6006. The process then retrieves the processed data from the compute device(s) and signals the Proof of Work Component that the processing is complete for this End Users Devices 6007.

[0053] FIG 7 is a flow diagram of the verification step that all data was really processed. This is also known as “Real Proof of Work”. The process first grabs the Transaction data from the Generic Blockchain 7001 as tdata. It then sends tdata to the Verification Step 7002. If the Verification step returns that all blocks are verified 7003, it then sends a message to the Generic Blockchain that the transaction has completed

successfully 7004. If all blocks have not been verified, the process then sends the bad blocks back to the Generic Blockchain for reprocessing 7005.

[0054] FIG 8 is a flow diagram of the verification step that verifies all blocks of a transaction request to process data. For each of the number of processing requests a random block is checked from the verification device. A verification device is random End User Device that is chosen that receives a block from each of the other End User Devices and runs the processing of those blocks for verification. A counter 'i' is set to 0 as the first block index to check 8001. A check is made for counter 'i' is less than the number of devices involved in the processing request. If yes, 'i' is used to index into a table of indexes that correspond to the random block that was chosen to verify of a device's processed data and that device is set to verified 8003. The checksum of the verification block is checked against the random block 8004. If the verification block checksum is equal to the system loops to checking the next device that processed data. If the verification fails, a table index is marked as false for the failed device 8005. The process continues to verify all the devices random verification blocks. If all the devices have been checked, the return value is an array of device verification pass or failure flags.

[0055] Different features, variations and multiple different embodiments have been shown and described with various details. What has been described in this application at times in terms of specific

embodiments is done for illustrative purposes only and without the intent to limit or suggest that what has been conceived is only one particular embodiment or specific embodiments. It is to be understood that this disclosure is not limited to any single specific embodiments or enumerated variations. Many modifications, variations and other embodiments will come to mind of those skilled in the art, and which are intended to be and are in fact covered by both this disclosure. It is indeed intended that the scope of this disclosure should be determined by a proper legal interpretation and construction of the disclosure, including equivalents, as understood by those of skill in the art relying upon the complete disclosure present at the time of filing.

[0056] Although the instant invention has been described in relation to particular embodiments thereof, many other variations and modifications and other uses will become apparent to those skilled in the art.

ABSTRACT:

[0057] A system and method for distributed general-purpose computing with cryptotoken payment system provides a full system that integrates a slow Blockchain Cryptotoken with a fast Directed Acyclic Graph Blockchain Cryptotoken. The instant invention provides: (1) a hybrid Cryptocurrency/Cryptotoken system that allows for very fast transactions to happen decoupled from the slow transactions of normal Cryptocurrencies (2) a mechanism to verify the processing of data and payments to the End User's wallet for the processing of that data that can in turn be used for In App purchases or in game purchases of items and (3) mechanisms for the transfer of unprocessed and processed data to the client or customer.

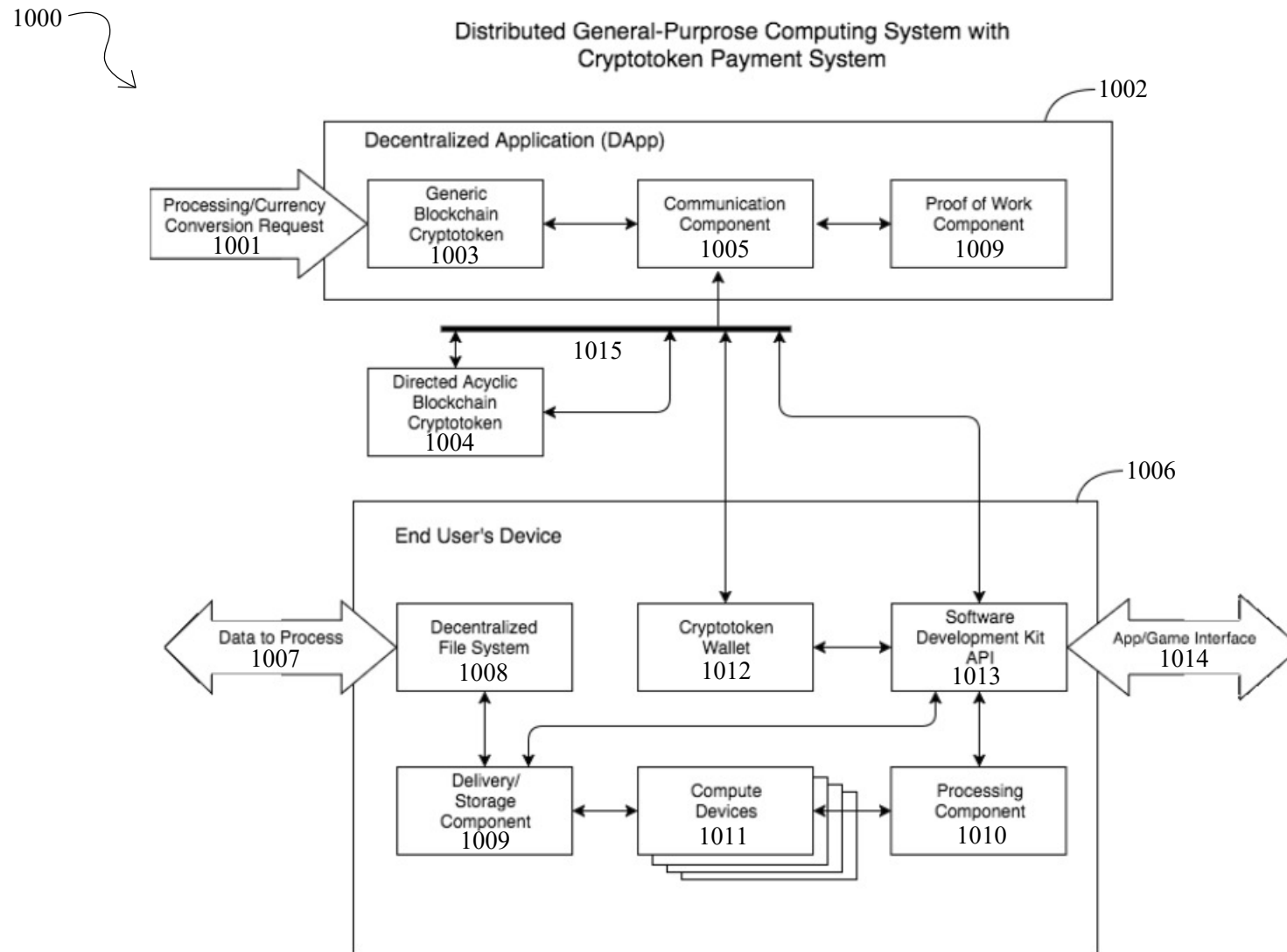


FIG. 1

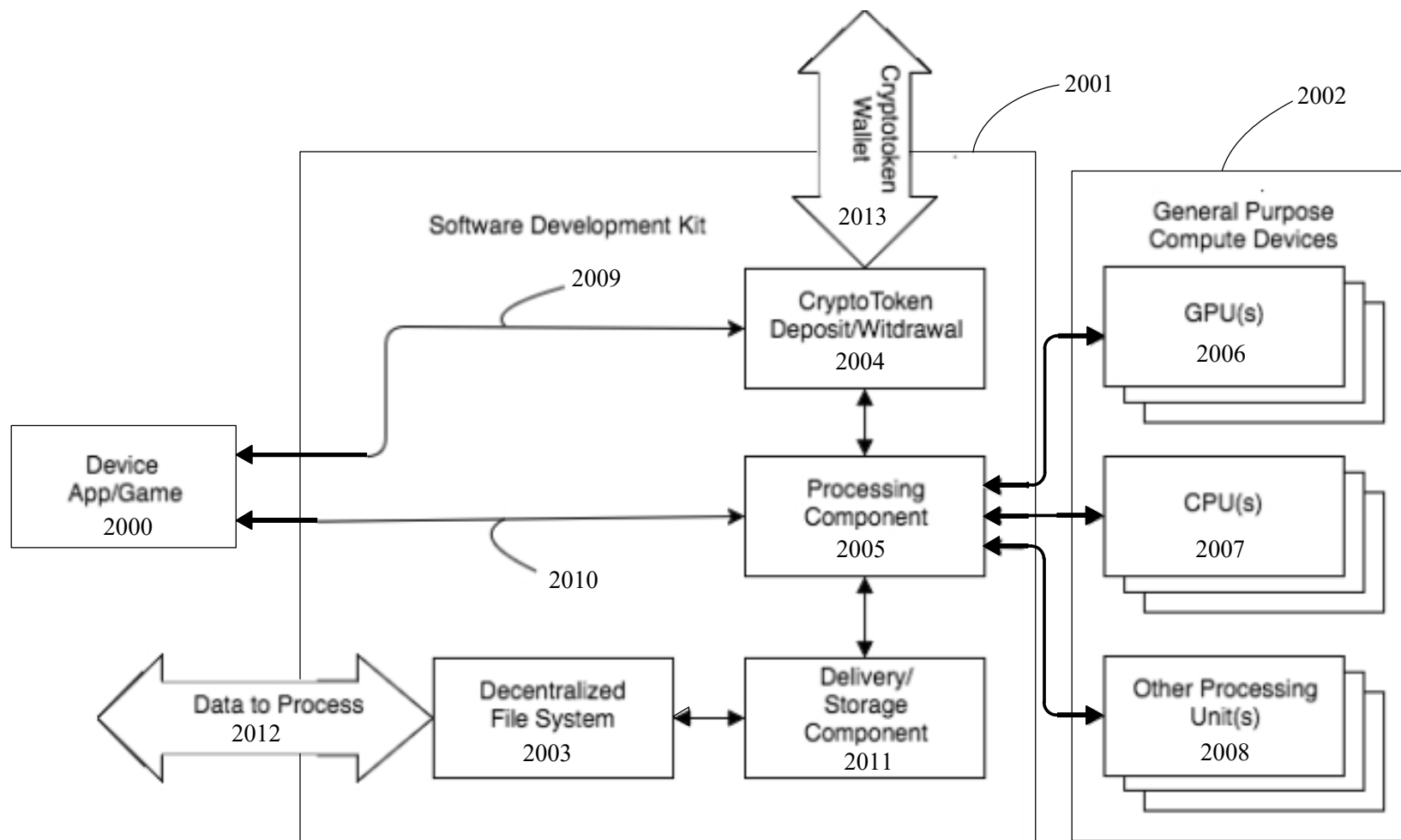
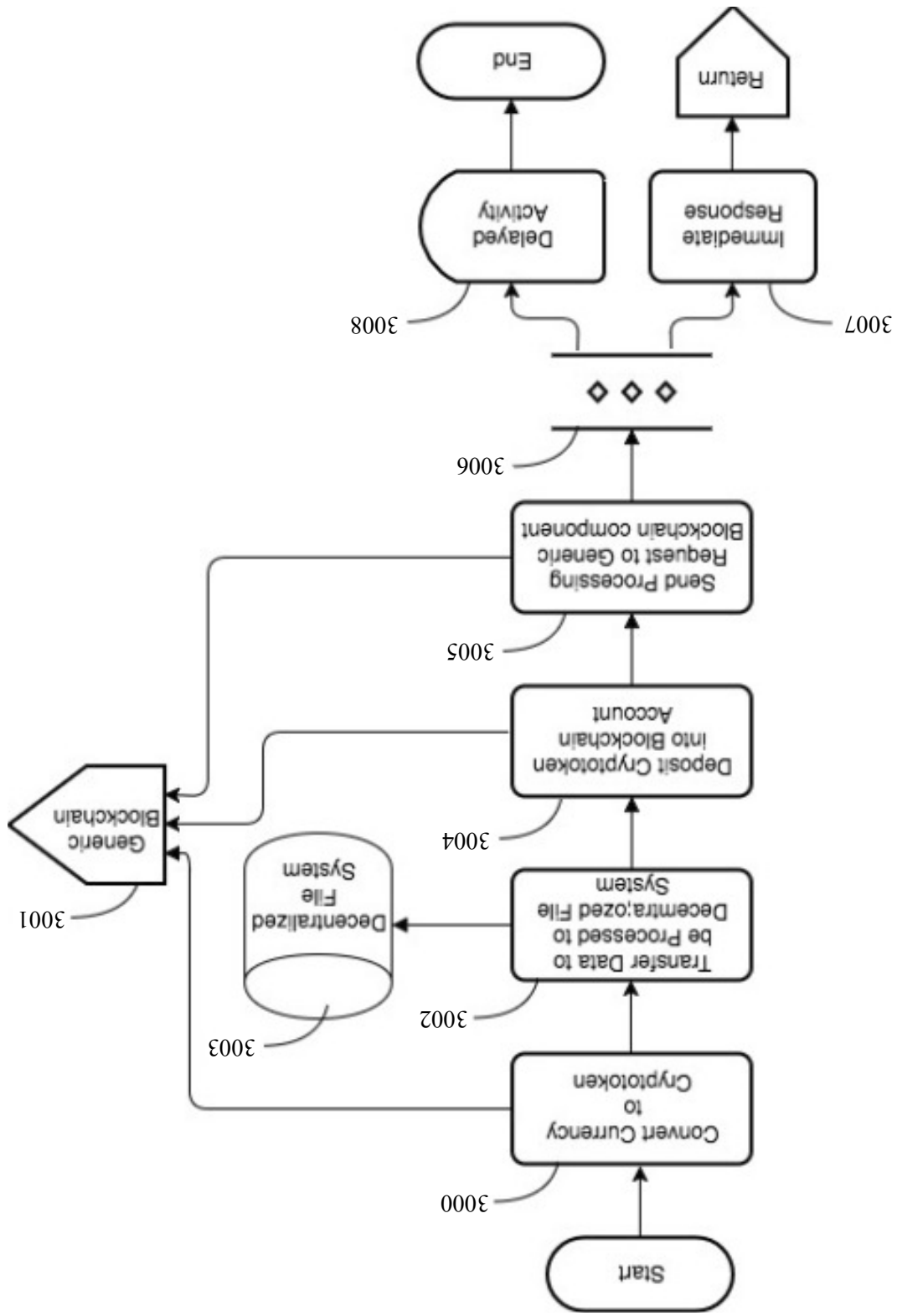


FIG. 2

FIG. 3



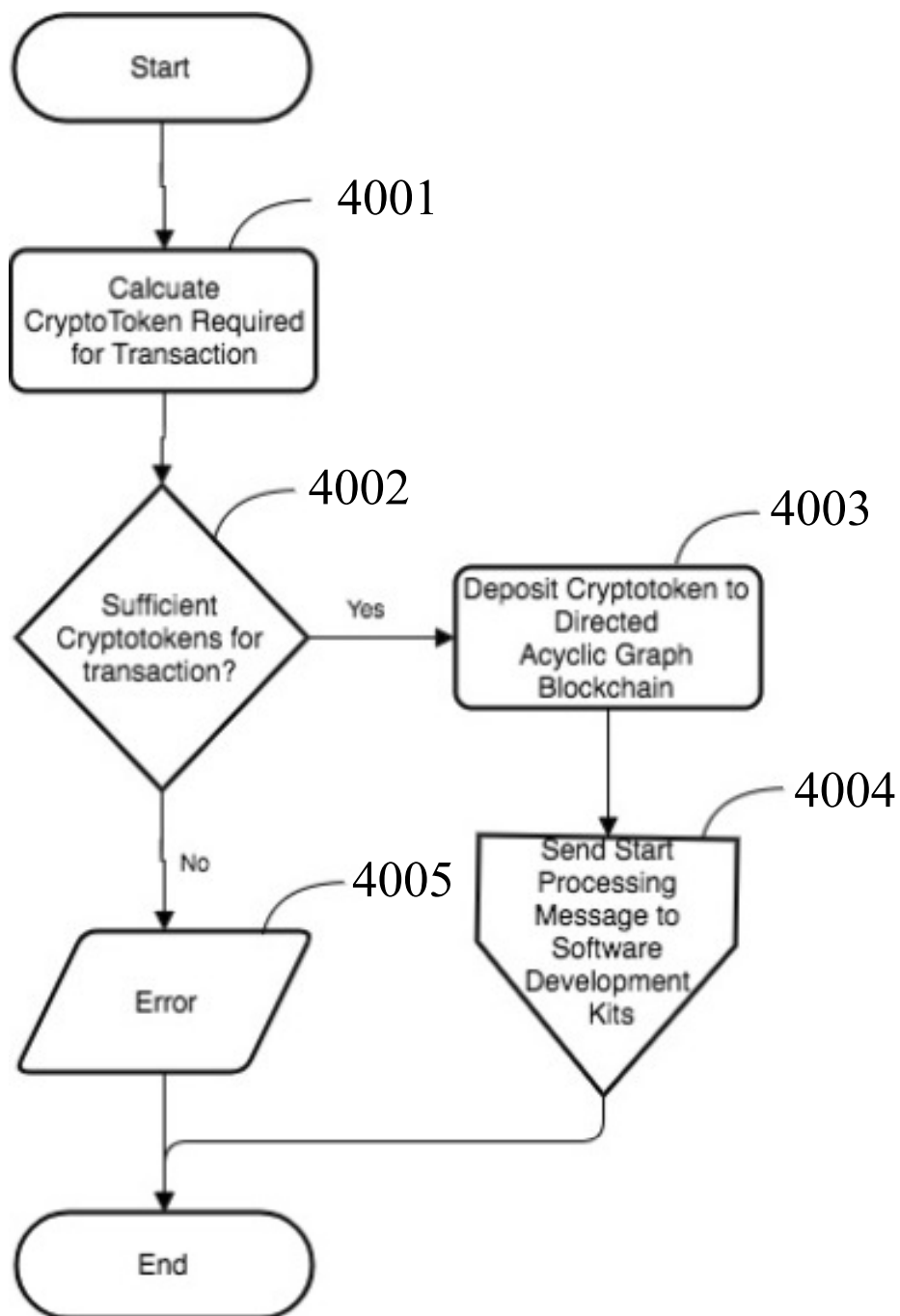


FIG. 4

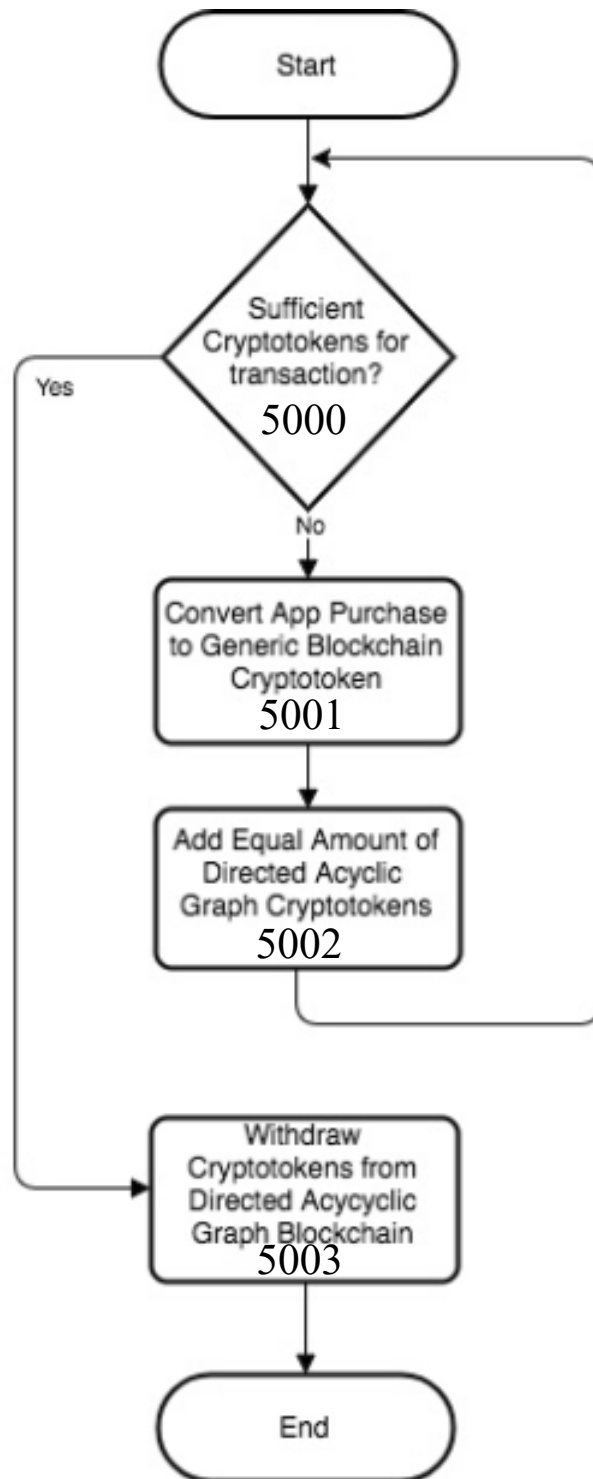


FIG. 5

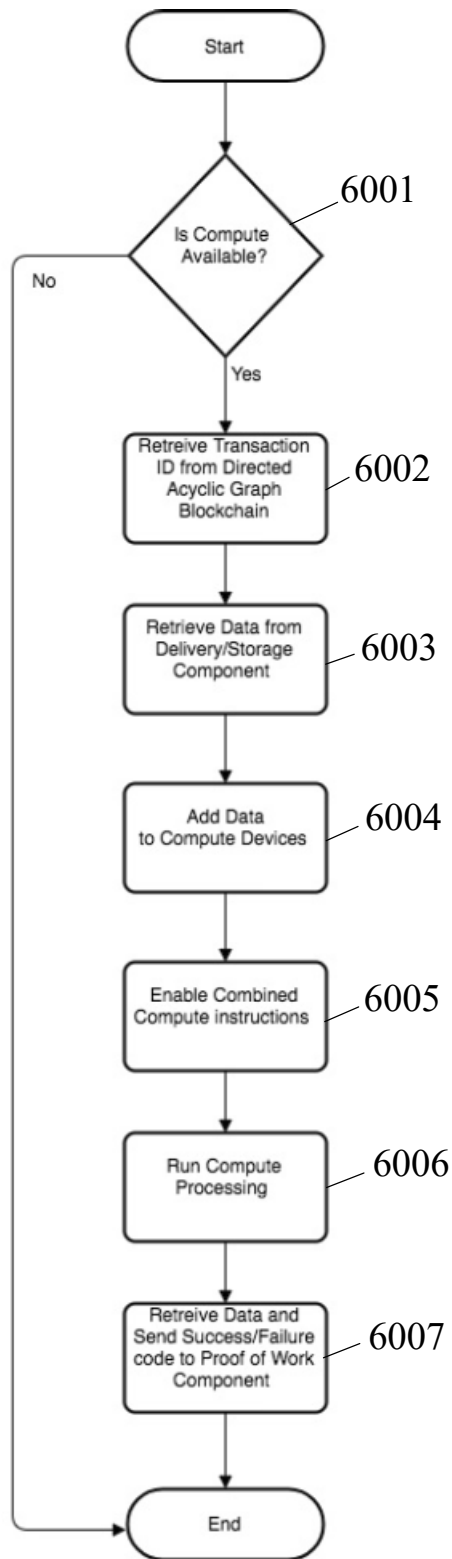


FIG. 6

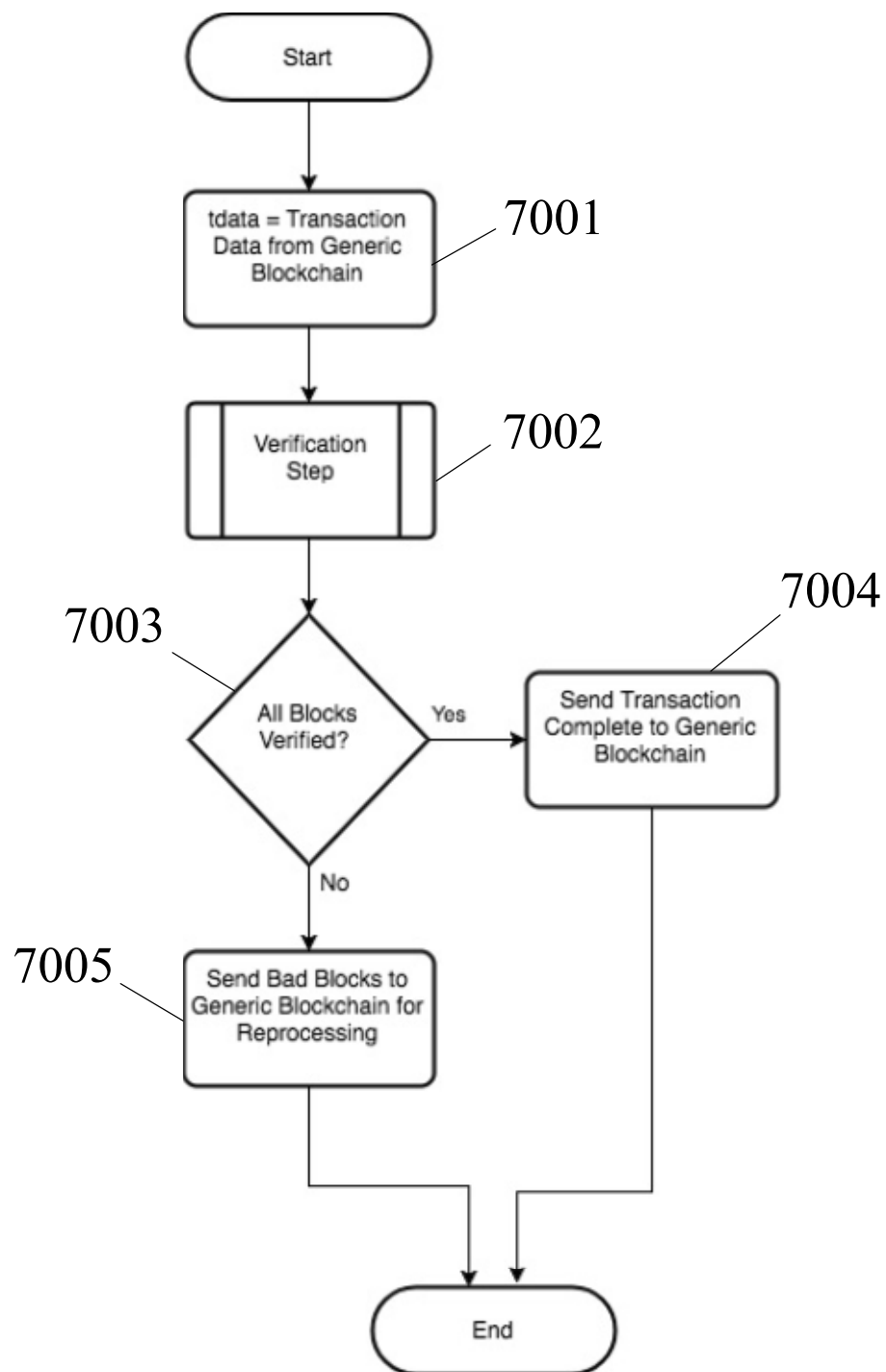


FIG. 7

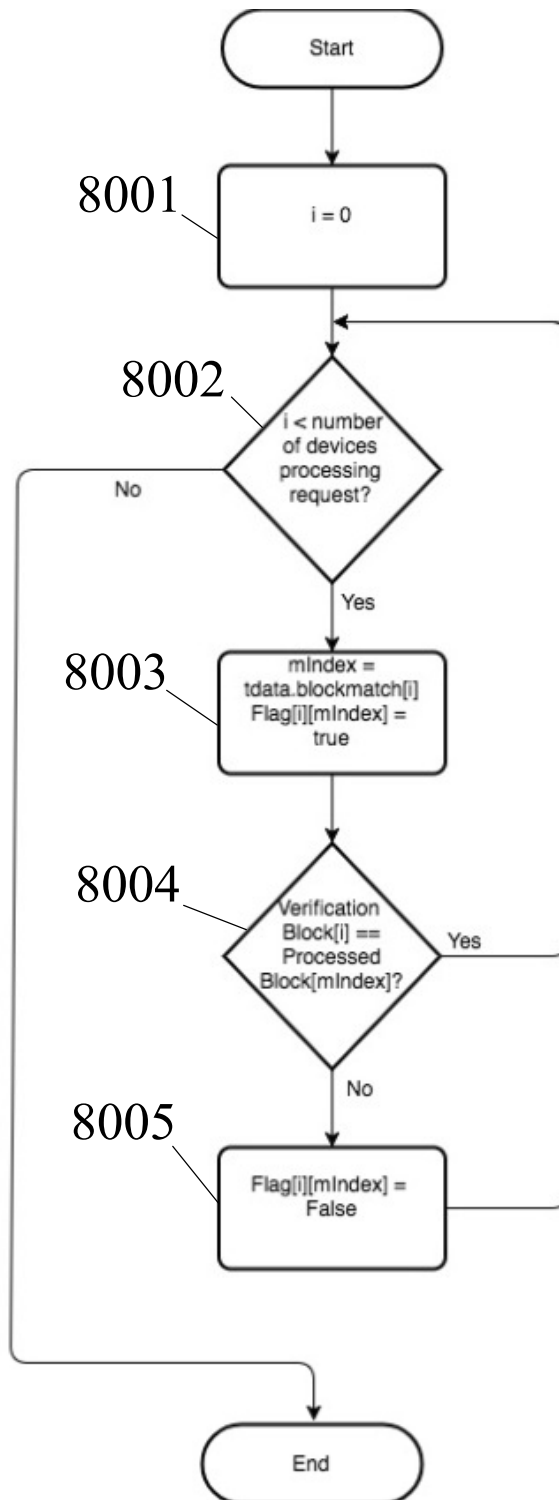


FIG. 8